AAROHAN INTER-CENTRE HACKATHON 2026

AAROGYA DRISHTI

Building an AI-Assisted Oral Cancer Early Detection,
Telemedicine and Patient Care Platform

Organised by: Integrated Tribal Development Agency (ITDA), Paderu

Alluri Sitharama Raju District, Andhra Pradesh

Participants: Students of AAROHAN AI and Software Engineering Training Centres

27 OCTOBER - 4 NOVEMBER 2026

AI for Public Health | Technology for Tribal Communities | From Screening to Treatment

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FIRST PRIZE: INR 10,000 | TEAM SIZE: MAXIMUM 5 STUDENTS

Official Hackathon Problem Statement, Technical Specifications, Competition Rules and Evaluation Framework

1. THE CHALLENGE

Can you build a technology platform that helps detect suspicious oral lesions early and ensures that patients in remote tribal villages receive timely medical care?

Oral cancer and oral potentially malignant disorders are significant public-health concerns in India. Early detection can be challenging in remote areas where specialist services are limited. Persistent mouth ulcers, white or red patches, abnormal growths and related symptoms may go unassessed. Frontline workers can conduct initial examinations, but specialist interpretation, referral and follow-up can be difficult to coordinate.

The challenge: Develop AAROGYA DRISHTI, a complete, functional, secure, AI-assisted digital platform connecting patients, village health workers, PHC Medical Officers, specialists, diagnostic facilities and district administrators within one continuous workflow.

The platform should support photographic documentation and appropriate AI-assisted review of suspicious abnormalities, clinical assessment, appointments, diagnostics, longitudinal medical records and treatment linkage. **Build a usable, deployable healthcare workflow - not just a website or a standalone image classifier.**

2. INSPIRATION: IISc TANUH - AAROGYA AAROHAN

Participants should study the Aarogya Aarohan oral-screening innovation developed by the Indian Institute of Science (IISc), Bengaluru, through its Translational Artificial Intelligence for Networked Universal Healthcare (TANUH) initiative. Publicly described concepts include mobile-phone oral photographs, AI-supported assessment, offline operation and specialist-supported screening.

- IISc TANUH: https://www.tanuh.ai/
- Oral cancer screening: https://www.tanuh.ai/solutions/oral-cancer
- Aarogya Aarohan application: https://play.google.com/store/apps/details?id=in.ac.iisc.arogyam
- IISc: https://www.iisc.ac.in/

Intellectual-property and clinical boundary: Do not copy proprietary code, imply endorsement or claim access to IISc models. Any real clinical integration requires formal permission, suitable technical interfaces and required medical-device and clinical approvals. Aarogya Drishti must extend beyond screening into the complete referral-to-treatment pathway.

3. EXPECTED FINAL PRODUCT

1. **Health Worker Mobile App:** registration, history, examination, photos, offline records and synchronization.
2. **AI-Assisted Screening:** image quality, transparent demonstration inference, appropriate visual outputs and limitations.
3. **Medical Officer and Specialist Portal:** review, clinical observations, decisions and referrals.
4. **Telemedicine:** secure video consultation and documentation.
5. **Appointment and Referral:** availability, booking, reminders and completion tracking.
6. **Diagnostics and Treatment:** investigation orders, biopsy/pathology, medically indicated imaging and follow-up.
7. **Patient Record Management:** secure longitudinal case history and audit log.
8. **District Dashboard:** PHC/mandal summaries and case-management indicators.

The result must be responsive, usable in Telugu and English, and demonstrated on an ordinary Android phone and computer.

4. DETAILED FUNCTIONAL REQUIREMENTS

Module A. Patient Registration and Medical History

- Application-generated unique patient identifier; name, age/date of birth, sex, contact details.
- Village, habitation, Gram Panchayat, mandal and PHC details.
- Exposure history: smoking, chewing tobacco, gutkha, areca nut, reverse smoking, relevant habits, duration and frequency.
- Symptoms: persistent ulcers, white/red patches, pain, swallowing difficulty, restricted mouth opening, bleeding or visible growth; symptom duration.
- Past screening, relevant history and referral history.
- Appropriate consent records for examination, imaging, teleconsultation and data processing.
- Duplicate detection; longitudinal record retrieval; Aadhaar not mandatory.

Module B. Smartphone Oral Screening

- In-app camera activation and guided, clinically appropriate oral views of tongue, cheeks, palate, gums and floor of mouth.
- On-screen framing and lighting guides, image preview, recapture and body-site annotation.
- Blur/darkness/glare checks; images linked securely to individual case.
- Encrypted offline saving and safe synchronization; prohibit unauthorized photo sharing.
- Emphasize that photographs supplement, and never replace, the full clinical oral examination.

Module C. AI-Assisted Abnormality Assessment

- Build a modular inference interface to accept images, handle poor quality and return structured findings.
- Where the model genuinely supports localization, show image overlay, bounding box or heatmap; never invent localization.
- Demonstration states: image insufficient; suspicious finding for clinical review; no suspicious finding identified by demonstration model.
- Display model provenance, version, permitted use and clear NOT FOR CLINICAL USE labeling for research/mock models.
- Medical Officer may agree, disagree or defer; store model output separately from signed clinical conclusion.
- Track model version, timestamp and history of all outputs.

Non-negotiable: Student-trained or public-image models are not validated cancer detectors. The app must never independently confirm or stage cancer, or assure a patient they are cancer-free. Scores reward transparent integration and responsible engineering, not unsupported accuracy claims.

Module D. Medical Officer and Specialist Review

- Secure queue of submitted cases; clinical history, oral photos, zoom and advisory AI output.
- Record findings, request recapture or more information, and document clinical decisions.
- Demonstrate routine follow-up, specialist review and urgent-referral workflow using approved sample rules.
- Generate referral; record reasons; timestamp and sign clinical notes.
- Separate specialist access and consultation notes; qualified clinicians retain final responsibility.

Module E. Telemedicine and Video Consultation

- Specialist consultation request, available slots, booking, rescheduling and cancellation.
- Secure time-limited meeting link, working audio/video and restricted access.
- Show relevant cases to authorized participants; capture date, clinician, notes, outcome and next action.
- Notify patient/worker without revealing sensitive details; disable video recording by default.
- Use an appropriate video SDK and explain how telemedicine compliance would be met before real deployment.

Module F. Physical Hospital Appointment and Referral

- Choose specialty and designated hospital; record referral reason and priority.
- Request/book appointment; authorized hospital confirmation, rescheduling and attendance.
- Record time, location, assigned service and printable Telugu/English referral slip.
- Reminders, overdue cases, missed-visit reasons and escalation to patient navigator.
- Track completion of the specialist assessment, not only referral creation.

Module G. Diagnostic Investigation and Confirmation

- Specialist-directed clinical re-examination, biopsy, histopathology, CT/MRI when indicated or other investigations.
- Tracking states: advised, scheduled, performed/sample collected, report pending, received, clinician reviewed, next action.
- Distinguish screen-positive abnormality, clinical suspicion, precancer and pathology-confirmed malignancy.
- A photograph or AI flag never substitutes for a clinically indicated biopsy and pathology.

Module H. Longitudinal Patient Record Management

- Single case timeline: registration, consents, screening visits, images, AI history, signed medical notes, video consultations and referrals.
- Investigation requests, reports, clinician-entered diagnosis, treatment linkage, follow-up and missed visits.
- Every material change must retain user identity, time, version and an auditable trail.

Module I. Patient Navigation and Follow-Up

- Queue of pending specialist assessments, investigations and treatment linkages.
- Assign navigator, contact attempts, transport support and follow-up tasks.
- Telugu/English non-sensitive reminders, overdue escalation, missed-appointment reasons and treatment initiation tracking.

Module J. District Administrative Dashboard

- Counts and rates for screenings, doctor reviews, suspicious findings, specialist referrals, completed appointments and overdue cases.
- Biopsies and pathology pending/completed, clinician-confirmed precancers/cancers, treatment linkage and loss to follow-up.
- Average time to review/diagnosis; village/PHC/mandal filters and aggregate export.
- Administrative users see aggregated programme indicators, not identifiable patient records by default.

5. TELUGU, ACCESSIBILITY AND LOW-CONNECTIVITY USE

Mandatory: Telugu and English for essential field workflows; one-touch switching; large icons and clear buttons; minimal typing; low-literacy-friendly screens; Telugu error messages; operation on budget Android devices; legible mobile layouts and accessible touch targets.

Desirable: Telugu voice guidance and verified voice-to-text, patient explanation screens, local-language adaptability and assistive accessibility features. Qualified Telugu-speaking clinicians should review medical translations before real use.

6. SECURITY, PRIVACY AND MEDICAL DATA PROTECTION

Authentication and access

- Individual secure accounts; role-based access; least privilege; strong sessions; MFA for privileged users.
- No shared accounts; prevent access to patient records outside an assigned care relationship.

Storage, communication and governance

- HTTPS/TLS, encrypted sensitive records at rest, secure photo/document storage and no public clinical-media links.
- Protect offline files and credentials; backup/recovery; safe error logging; prevent unauthorized bulk export.
- Log record access, edits and exports; retain AI versions, clinical corrections and audit evidence.
- Separate consent capture, defined retention policies and privacy-by-design planning for Indian health-data/DPDP requirements.

Hackathon restriction: Synthetic patient records, actors and licensed educational/demo images only. No actual patient records, photographs or clinical video consultations may be collected in the competition. Any live-data field pilot later requires health-authority authorization, clinical protocols, relevant regulatory/ethics permissions and independent security review.

7. TECHNICAL EXPECTATIONS

Component	Illustrative technology
Mobile	React Native / Flutter
Web	React / Next.js
Backend	FastAPI / Node.js
Database	PostgreSQL
AI	Python service or permitted external API
Image storage	Encrypted object storage
Access control	Role-based authorization
Video	Secure video SDK or suitable service
Deployment	Containerized/approved cloud environment
Monitoring	Structured logs, health checks, error monitoring

Mandatory engineering evidence: deployed and accessible working demo; end-to-end real backend/data flow; persistent database; authenticated roles; validation/error handling; functioning photos, appointments, referrals and one video session; offline demonstration; secured APIs; Telugu workflow; transparent clinical/AI separation; operational documentation. A static prototype does not qualify.

8. COMPULSORY END-TO-END DEMONSTRATION

Demo case: A 46-year-old synthetic patient from G. Madugula has a persistent white cheek patch and a tobacco-chewing history.

1. Register patient in Telugu and record relevant symptoms and exposure.
2. Capture/upload permitted demonstration oral photos and test image quality.
3. Process with a clearly labeled illustrative AI service; explain its limitations.
4. Medical Officer reviews and documents decision.
5. Book and demonstrate a secure specialist video consultation.
6. Generate a physical specialist appointment when required by scenario.
7. Upload sample biopsy order and simulated pathology report.
8. Authorized demo clinician records final simulated clinical result.
9. Update the longitudinal record, assign navigator and update district dashboard.
10. Demonstrate role restrictions, data privacy and audit trail.

Additional live jury tests: switch languages; offline registration and re-sync; unauthorized record access blocked; poor image handled; doctor override of AI; missed appointment; failed network request and safe recovery; clear separation of demo AI from diagnosis.

9. HACKATHON SCHEDULE: 27 OCTOBER - 4 NOVEMBER 2026

Date	Milestone
27 October	Launch, problem briefing, team registration
28 October	Clinical workflow, design and architecture
29 October	Registration, database and screening app
30 October	Image capture, quality checks and AI prototype
31 October	Medical Officer and specialist portal
1 November	Video call, booking and referral module
2 November	Diagnostics, records, Telugu and security
3 November	End-to-end testing, deployment, semifinal review
4 November	Final live demos, jury scoring and awards

Teams must use version control, record meaningful contributions and maintain a daily development log.

10. JUDGING AND EVALUATION

Criterion	Marks
Complete workflow and integration	20
AI engineering, image-quality and safe integration	15
Doctor review, teleconsultation and appointments	15
Records, diagnostics and navigation	15
Security, privacy and governance	15
Telugu, offline use and accessibility	10
Deployment, testing and documentation	10
TOTAL	100

Minimum eligibility for awards: deployed demo; functioning backend/database and access control; Telugu essential workflow; real review/referral and record management interactions; one operational demonstration video call; synthetic/authorized demo data; honest labeling of AI simulation.

11. FINAL SUBMISSION REQUIREMENTS

1. Version-controlled source repository and disclosure of third-party code/AI tools.
2. Application package/build and deployed demo address.
3. Evaluator test accounts and instructions.

4. Architecture diagram and database schema.
5. API documentation and installation/deployment guide.
6. AI model/service provenance, version, integration diagram, limitations and validation approach.
7. Synthetic demonstration dataset and reset instructions.
8. Security/privacy design and role-access matrix.
9. Test report including failure cases and offline tests.
10. Telugu user-interface demonstration.
11. Five-minute product demo video.
12. Live end-to-end jury presentation.
13. Roadmap from demonstration to approved clinical pilot.

12. LONG-TERM DEPLOYMENT VISION

The most promising submission may serve as a starting codebase for a mentored district-health project, subject to independent review. Possible future extensions are a formal clinical AI partnership with IISc TANUH, alignment with NP-NCD workflows, a supervised PHC pilot, expansion to additional mandals and a separate cervical-screening referral module. Winning a hackathon does not confer authorization to conduct clinical screening or process real medical data.

13. MESSAGE TO PARTICIPANTS

This is not simply an exercise in Python, React, databases or AI. Build a system that allows a health worker to document a screening, a clinician to review suspicion, a specialist to provide timely care, and the district to ensure no patient is lost between referral and treatment.

Build software that works. Build an application that people can actually use. Build a system that protects patient information. Build technology that improves access to healthcare.

Grand Challenge: "From village-level oral examination to timely specialist diagnosis and treatment - build the complete digital pathway."

14. COMPETITION RULES, ELIGIBILITY AND PRIZE

14.1 Participation and team formation

- Only currently enrolled AAROHAN students from participating centres are eligible; mentors/coordinators cannot compete as team members.
- Teams must contain **1 to 5 students maximum**. No sixth member is permitted.
- A student may join only one team. Cross-centre teams are allowed if all members are eligible, subject to coordinator verification.
- Teams must nominate one leader and disclose member names, centre, assigned roles and repository before coding begins.
- Roster must be finalized at registration on 27 October. Substitutions require organizer approval for documented exceptional reasons.

14.2 Originality and permitted tools

- Open-source libraries, AI coding assistants, frameworks and properly licensed datasets may be used with attribution and license compliance.
- The substantial project integration and original implementation must be carried out during the hackathon; reusable pre-existing components must be declared.

- No plagiarism, stolen code, purchased finished projects, fabricated demonstrations or falsely claimed original AI models.
- Mentors may guide and review but may not write the competition submission on behalf of a team.

14.3 Conduct, checkpoints and final submission

- Teams participate during 27 October to 4 November 2026, follow organizer checkpoints and maintain daily commits.
- Final code and documentation must be frozen by **10:00 AM IST on 4 November 2026**; only fixes expressly permitted by the jury may follow.
- Teams must demonstrate the app live using supplied synthetic scenarios and permit technical questioning and spot checks.
- Internet-dependent demos must disclose dependencies; evaluators may simulate connectivity loss and data-access attacks.
- Respectful collaboration and protection of organizer equipment and credentials are mandatory.

14.4 Prize and selection

FIRST PRIZE: INR 10,000 (Rupees Ten Thousand only) for the overall winning team, awarded as a single team prize, not INR 10,000 per participant. The winning team decides its internal distribution by mutual agreement. Certificates of recognition may be awarded to finalists and other deserving teams subject to organizer decision.

Winner determined using the 100-mark rubric above. In case of a tie, higher scores in (1) patient workflow, then (2) security/privacy, then (3) AI safety/integration will decide; remaining ties will be resolved by a documented jury decision. The jury may withhold the prize if no project satisfies mandatory safety and functionality criteria. Award payment is subject to applicable administrative procedures.

14.5 Ownership, deployment and disqualification

- Teams retain rights in original code they create, subject to underlying open-source and third-party licenses. Any later transfer, public release or government deployment must be separately agreed in writing; winning alone does not automatically transfer intellectual property.
- All teams must submit reproducible source and a permission to evaluate it for this hackathon.
- Organizers may disqualify teams for using real patient data, unauthorized clinical claims, exposed health records, plagiarism, serious misconduct, or demonstrably falsified results.
- A basic safety violation may disqualify a team notwithstanding high technical scores.
- Organizer/jury decisions on procedural matters are final, subject to fair review of documented scoring or eligibility errors.

14.6 Clinical safety disclaimer

All competition applications, demonstrations, predictions, images, consultations, referrals and pathology examples are **SIMULATIONS - NOT FOR CLINICAL USE**. No student team may clinically screen, advise, reassure, diagnose or treat a real patient on the basis of this platform during the hackathon. Any production use requires clinical governance, privacy and security review, and applicable permissions.

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